

PROJECT PROPOSAL

GROUP PROJECT

CST 394-2

Group No: CST 07

**Smart Agricultural Decision Support System
With Risk-Aware Recommendations**

Computer Science and Technology

Department of Computer Science and Informatics

Faculty of Applied Sciences

Uva Wellassa University of Sri Lanka

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DECLARATION

We hereby declare that the project will be developed by us and will be our own effort and that no part will be plagiarized without citations under the supervision of Ms. D.M.A.Sewwandi, Lecture, Department of Computer Science and Informatics, Faculty of Applied Science, Uva Wellassa University of Sri Lanka. This Project Proposal is submitted for the partial fulfillment of the requirement of the course unit CST 394-2, Project II for the degree of Computer Science and Technology.

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Chapter 1: Introduction

1.1 Project Title

Smart Agricultural Decision Support System with Risk-Aware Recommendations for Sri Lanka

1.2 Project Description

The Smart Agribusiness Decision Support System is a web-based platform designed to assist land owners and farmers, small-scale cultivators, individuals entering agriculture without prior farming experience. The system integrates farm management, machine learning based crop recommendations with Explainable AI and risk_aware guidance and a digital marketplace to support decision- making and business operations. Users can either manage farming activities by themselves or supervise farmers with the support of the system.

The platform focuses on key functionalities such as machine learning-based crop recommendations with explanations generated through Explainable AI techniques risk-aware suggestions, rule-based crop guidance, formula-based yield estimation, selling price calculation, and a marketplace where users can connect with traders. The system provides a simple and user-friendly interface to manage farm activities, monitor performance, and handle buying and selling processes.

This initial version will include machine learning-based crop recommendations with Explainable AI explanations using trained machine learning model based on agricultural datasets, and essential CRUD operations for managing crops, users, and transactions. The system aims to provide a strong foundation that can be extended with advanced machine learning capabilities and real- time data integration in future versions.

1.3 Background And motivation

Agriculture remains one of the most important sectors in Sri Lanka's economy. However, many individuals, especially newer generations, face difficulties when entering agriculture due to lack of knowledge about soil conditions, crop selection, weather patterns, and risk factors.

At present, most farming decisions are made based on experience or traditional knowledge, which may not always lead to optimal results. Additionally, there is a lack of digital platforms that connect producers directly with buyers, resulting in inefficiencies in the supply chain.

Many new entrepreneurs are interested in agriculture as a side business but lack access to proper guidance and decision support tools. They also face challenges in predicting profitability and managing farm operations efficiently.

Therefore, there is a strong need for a system that:

- Provides guidance on crop selection and cultivation with explanations
- Offer risk-aware suggestions
- Estimates yield and selling prices
- Connects land owners with traders
- Digitizes farm management processes

1.4 Problem in Brief

Despite the importance of agriculture, several key problems exist:

1. **Lack of Knowledge:** Beginners do not have sufficient knowledge about crops, watering/fertilizer conditions, pest/disease risks.
2. **Poor Decision Making:** Beginners select crops without proper analysis, leading to losses.
3. **Limited Estimation Systems:** There are limited tools to estimate yield before cultivation and difficulties in estimating profitable selling prices.

4. **Market Inefficiency:** Beginners struggle to find buyers, and traders lack direct access to producers.
5. **Manual Management:** Farm activities are mostly managed manually without digital support.

1.5 Proposed Solution

The proposed system addresses these issues through the following features:

1. **Crop Recommendation and Guidance System:** Suggests suitable crops using a machine learning model trained on agricultural data sets with Explainable AI support. After a crop is selected, the system provides rule-based cultivation guidance such as planting instructions, watering schedules, fertilizer application guidance, pest/disease warnings, and climate-related alerts.
2. **Yield and Selling Price Estimation Module:** Estimates crop yield and calculates selling price using mathematical formulas based on land size, production factors, and cost of cultivation. The selling price is determined by adding a desired profit margin to the total production cost.
3. **Market place Integration:** Enables crop buying and selling between managers and traders, with basic negotiation and transaction communication features between land owners and traders.
4. **Farm Management System:** Enables users to manage farm data, crop status, and activities.
5. **Role-Based Access Control:** Different user roles such as Admin, Land Owner, and Trader ensure structured system usage.
6. **Smart Agricultural Calendar Tracking System** The system enables land owners to schedule, manage, and monitor farming activities such as planting, fertilization, irrigation, and harvesting. It organizes farm operations in a calendar view to ensure timely and efficient execution based on farm requirements and conditions.

1.6 Project Aim and Objectives

Aim:

To develop a user-friendly web-based system that supports agribusiness decision-making with risk-aware recommendations, improves productivity, and connects producers with buyers in Sri Lanka.

Objectives:

1. To develop a Machine Learning-based crop recommendation system using environmental and agricultural inputs with Explainable AI (XAI) techniques to provide accurate and understandable crop recommendations.
2. To implement a rule-based crop guidance system with risk-aware suggestions, pest and disease warnings, and climate-related advisories to support informed agricultural decision-making.
3. To implement formula-based crop yield estimation and selling price calculation using land size, seed quantity, production cost, and profit margin.
4. To build a farm and crop management module enabling land owners to manage farm data, crop records, and activity tracking.
5. To design a digital marketplace connecting sellers and traders for crop buying and selling through a negotiation system.
6. To implement a secure role-based access control system for all user role

Chapter 2: Methodology

Agile methodology is well-suited for the development of the system as it allows for flexibility, iterative development, and continuous improvement. Here's an overview of the methodology and the process followed throughout the software lifecycle.

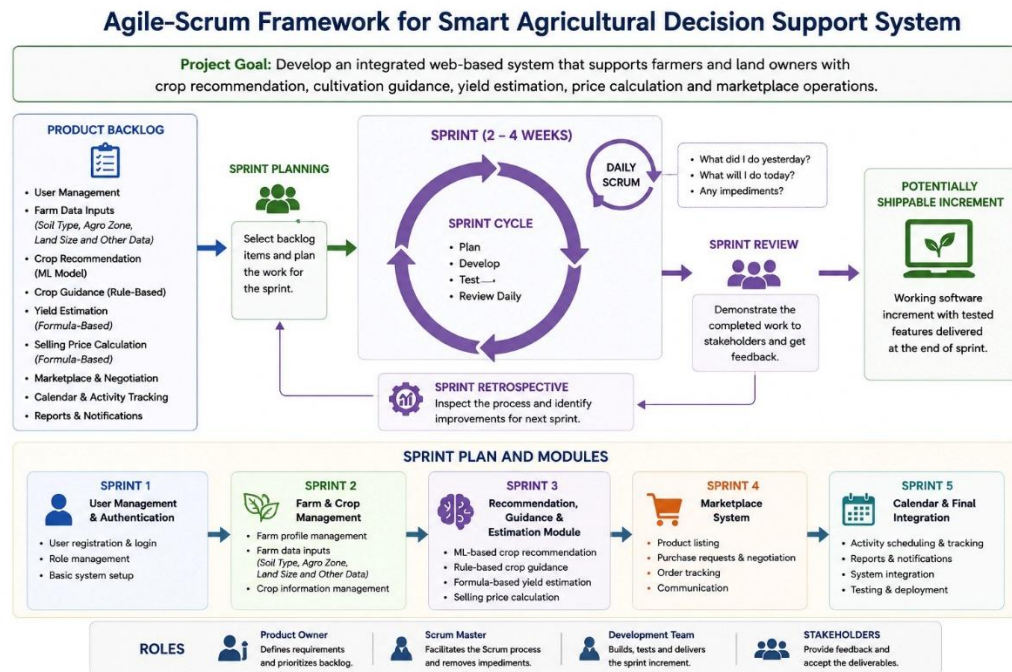


Figure 1: The Agile-Scrum Framework

Sprint 1: Core System Architecture and User Management

1. Project Initiation and Release Planning:

- Define the scope of core system functionalities
- Identify primary user roles (Visitor, Admin, Land Owner, Trader)
- Prioritize essential features for system setup and user management

2. Sprint Planning:

- Plan system setup using FastAPI backend, React frontend, and postgresql database
- Design database architecture for users, farms, crops, and transactions
- Plan authentication system and role-based access control

3. Sprint Execution:

- Develop scalable database schema
- Implement core API endpoints
- Implement role-based access control for all user roles
- Develop user registration, login, and profile management
- Create basic account management for Admin, Land Owner, and Trader

4. Sprint Review:

- Demonstrate system architecture and database design
- Review authentication and access control
- Test user registration and login functionality
- Gather feedback on user management features

Sprint 2: Farm and Crop Management

1. Project Initiation and Release Planning:

- Define features for farm and crop management
- Map functionalities to land owner role
- Establish workflows for managing farm data and crops

2. Sprint Planning:

- Design farm management interface
- Plan crop creation, updating, and tracking features
- Plan storage of environmental data (soil type, season, etc.)

3. Sprint Execution:

- Implement farm data management (location, size, soil type)
- Develop CRUD operations for crops
- Create crop listing and updating features
- Build interfaces for monitoring crop status
- Implement basic data validation

4. Sprint Review:

- Demonstrate farm and crop management features
- Test CRUD operations and workflows
- Review usability of interfaces
- Gather feedback for improvements

Sprint 3: Crop Recommendation, Guidance and Estimation Module

1. Project Initiation and Release Planning:

- Define requirements for a Sri Lanka-specific crop recommendations with explanations, cultivation guidance, yield estimation, and selling price calculation system.
- Identify environmental and agricultural input parameters such as soil type,

Yala/Maha season, rainfall, humidity, temperature, and NPK values for accurate crop prediction and risk-aware recommendations

- Collect and prepare locally sourced agricultural datasets from the Department of Agriculture, Sri Lanka
- Select suitable machine learning classification models such as Random Forest and Decision Tree for crop recommendation and explanation generation.
- Define evaluation metrics including Accuracy, Precision, Recall, F1-Score, and Confusion Matrix for model validation
- Plan integration of crop recommendations, explanations, rule-based guidance, risk aware advisories, formula-based yield estimation, selling price calculation, farm management, and marketplace modules.

2. Sprint Planning:

- Design user interfaces for entering environmental and farm-related data.
- Plan machine learning workflows for crop recommendations with Explainable AI explanations, rule based crop guidance, risk_aware suggestions formula based yield and selling price estimation.
- Design recommendation logic tailored to Sri Lankan agricultural conditions.
- Plan backend APIs and frontend integration for recommendation, explanation generation, guidance, and estimation features.
- Define testing strategies for model accuracy, usability, and system performance.

3. Sprint Execution:

- Implement machine learning-based crop recommendation using agricultural datasets with Explainable AI explanations and rule-based crop guidance using predefined agricultural rules.
- Develop formula-based yield estimation and selling price calculation

modules using land size, seed quantity, seed fertility percentage, production cost, and profit margin.

- Train and test Random Forest and Decision Tree models for crop classification.
- Evaluate model performance using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
- Integrate recommendation, explanation generation, guidance, and estimation APIs with the frontend system.
- Display recommendations, estimations, and analytical insights in a user-friendly dashboard.
- Implement integrated farm management and marketplace functionalities.

4. Sprint Review:

- Demonstrate the machine learning-based crop recommendation, rule-based crop guidance, formula-based yield estimation, and selling price calculation features.
- Review recommendation accuracy and classification performance using evaluation metrics.
- Test system usability and functionality with sample agricultural datasets.
- Validate recommendations according to Sri Lankan farming conditions.
- Gather stakeholder and user feedback for model optimization and future improvements.

Sprint 4: Marketplace System

1. Project Initiation and Release Planning:

- Define marketplace requirements
- Identify workflows for selling and buying crops
- Map features to land owner and trader role

2. Sprint Planning:

- Design marketplace interface
- Plan selling, buying, and order tracking functionalities
- Plan transaction and communication flow

3. Sprint Execution:

- Implement crop selling functionality for land owners
- Develop trader purchase request system
- Create order management and tracking features
- Build transaction handling and status updates
- Implement basic notifications for users

4. Sprint Review:

- Demonstrate marketplace functionality
- Test buying and selling workflows
- Review order tracking system
- Gather feedback on user experience

Sprint 5: Smart Calendar and Final Integration

1. Project Initiation and Release Planning:

- Define requirements for smart calendar tracking and final system integration
- Plan calendar UI and activity scheduling workflows for land owners

2. Sprint Planning:

- Design the calendar interface and activity management module
- Plan integration testing strategy across all modules

3. Sprint Execution:

- Implement calendar-based tracking for farm activities (planting, fertilizing, watering, harvesting)
- Allow land owner to schedule, view, and update farming activities
- Conduct full system integration testing across all modules
- Perform UI/UX review, bug fixing, and final deployment preparation

4. Sprint Review:

- Demonstrate smart calendar and integrated system to stakeholders
- Conduct end-to-end system testing and gather final feedback
- Finalize deployment and complete project documentation.

Chapter 3: Requirements Identification

3.1 Functional and Non-functional requirements

3.1.1 Functional Requirements

1. Farm Management

- The system allows land owner to
 - ❖ add and manage farm details including location, size, and soil type
 - ❖ update and maintain farm information

- ❖ view all registered farms
- ❖ monitor farm status and activities
- The system stores environmental data related to farms such as soil type, season, rainfall, temperature, humidity, and other relevant agricultural inputs for crop recommendations with explanations, risk-aware suggestions and guidance.

2. Crop Management

- The system allows land owner to
 - ❖ add, update, delete, and manage crop records
 - ❖ track crop growth, status, and expected harvest dates
 - ❖ categorize crops based on soil type, season, and location
 - ❖ receive crop recommendations with explanations, risk-aware suggestions based on soil information, environmental conditions, and seasonal agricultural data
 - ❖ monitor crop suitability and cultivation risks for improved farm decision-making

3. Recommendation, Guidance, and Estimation Features

- The system
 - ❖ Provides crop recommendations with explanations by analyzing soil conditions, seasonal patterns, weather data, and environmental factors to support better farming decisions.
 - ❖ Estimates crop yield using predefined agricultural formulas and cultivation parameters.
 - ❖ Calculates suggested selling price based on estimated harvest quantity, production cost, and desired profit margin.
 - ❖ Displays estimation and recommendation with explanation results in a user-friendly format.

4. Marketplace

- The system allows
 - ❖ Land owner to display crops for sale
 - ❖ Traders to view available crops in the marketplace
 - ❖ Enables traders to send purchase requests and negotiate transaction details with land owner to accept or reject purchase requests
 - ❖ users to track order status (pending, accepted, rejected)
 - ❖ users to view transaction details

5. User Management

- The system
 - ❖ Allows Visitors to register and create user accounts
 - ❖ Provides authentication for all registered users
 - ❖ Allows Admin to assign and manage user roles (Admin, Land Owner, Trader)
 - ❖ Allows users to manage and update their profiles
 - ❖ Provides password reset and account recovery functionality
 - ❖ Tracks user activities within the system

3.1.2 Non – Functional Requirements

Performance:

- Efficient data retrieval for core functionalities
- Fast response time for user requests and system operations

Security:

- User authentication and authorization for all registered users
- Role-based access control for Admin, Land Owner and Trader

Usability:

- Intuitive and user-friendly interface
- Mobile-responsive design for basic user access

Scalability:

- Database structure that can accommodate future growth
- System architecture that supports expansion with additional features and integrations

3.2 System Requirements

3.2.1 Hardware Requirements

Development Environment:

- Minimum of 3–4 computers or laptops for the development team
- Each device should have at least 8GB RAM and a multi-core (preferably quad-core) processor
- Stable internet connection for development, testing, and integration activities

Production Environment:

- Web server with minimum specifications:
 - 8GB RAM
 - Multi-core processor
 - At least 250GB–500GB storage
- Database server with minimum specifications:
 - 8GB–16GB RAM
 - Multi-core processor
 - At least 500GB–1TB storage
- Reliable internet connection to support multiple users
- Basic backup solution to ensure data safety and recovery

3.2.2 Software Requirements**Development Tools:**

- Integrated Development Environment: VS Code
- Version Control System: GitHub
- API Testing Tool: Postman

Client Environment:

- Web browsers (Chrome, Firefox, Safari, Edge) for accessing the system

3.2.3 Technologies**Front-end:**

- React.js
- Tailwind CSS

Back-end:

- FastAPI(Python)
- RESTful API architecture

Database:

- postgresql database

Deployment:

- Cloud deployment on Render

3.3 User Roles

1. Visitor

Access Level: Public (No Login Required)

Features:

- Use AI-based crop recommendation with explanations feature
- Enter basic inputs (soil type, season, rainfall)
- Get suggested crops
- View basic agricultural knowledge (soil, crops, farming tips)
- Learn about farming before starting

2. Land Owner

Access Level: Registered User

Features:

- Manage farm details (location, size, soil type)
- Add, update, and delete crops
- Get AI-based crop recommendations with explanations and guidance
- View yield and selling price estimations

- Add crops for sale with price and quantity
- View trader requests
- Accept or reject purchase request
- Manage orders and delivery status

3. Trader

Access Level: Registered User

Features:

- View available crops for sale
- Search and filter crops
- View crop details (price, quantity, land owner info)
- Send purchase requests and engage in negotiation with land owner.
- Track order status (Pending, Accepted, Delivered)

4. Admin

Access Level: Full System Control

Features:

- Manage all users (create, update, delete)
- Assign roles (Land Owner, Trader)
- Monitor system activity
- Manage crop and system data
- Ensure system security and performance

Chapter 4: Project Plan (Gantt Chart)

Task	Week														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Client Interaction & Requirement Refinement															
Initial Planning & System Setup															
UI/UX Design (Iterative)															
Core Development (Modules)															
Continuous Testing & Debugging															
Feature Enhancements (Based on Feedback)															
Deployment Preparation															
Final Deployment & Maintenance															
Documentation															

Figure 2: Gantt Chart Representing the Project Timeline

Chapter 5: References

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